

Conditional Severity of Cascading Transmission Outages Under a Strictly Causal, Memory-Bearing Projection of Outage Streams

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Abstract

Forecasting whether cascading outages will occur in transmission systems is known to be difficult, and simple recent-activity baselines are hard to beat. This study asks a different question: given that a cascade occurs, can strictly causal information available beforehand identify periods in which it is more likely to become severe? We introduce a projection of the outage stream onto a small set of dimensionless state variables — a causal decoupling measure between observed and expected outage activity, a historically weighted (non-Markovian) accumulator of that decoupling, and an endogenous threshold that contracts with accumulated excess — and define flagged periods as intervals in which the system operates with positive but shrinking margin.

Applied to 14,258 automatic transmission outages from the Bonneville Power Administration (BPA) public record (1999–2017), the method yields a conditional probability of a large cascade (size ≥ 4 outages, 1-hour grouping) of 0.096 inside flagged periods versus 0.003 outside — an enrichment ratio of ≈ 28.8 (label-shuffle permutation test, $p < 0.001$; null 95th percentile ≈ 1.17). Under a matched alert budget, a causal trailing-intensity baseline reaches ≈ 3.2 . Partitioning cascades by signal overlap shows that intervals flagged only by the projection retain a conditional probability of 0.086, indicating information beyond recent outage intensity. For occurrence forecasting (6–48 h horizons), the trailing-intensity baseline remains superior, consistent with a record plausibly dominated by exogenous weather. As an external consistency check, the pipeline recovers a Zipf exponent of ≈ 2.99 for cascade sizes, close to previously reported values for this dataset.

On lower-density processed NYISO data (2008–2021), the same pipeline yields a weaker but significant enrichment (≈ 2.44 versus a permutation null of ≈ 1.26). An ablation replacing the causal decoupling measure with raw normalized intensity reproduces a measurement-degeneration failure discovered during development of the NYISO analysis (enrichment inverts to ≈ 0.55), isolating the source of that failure to the observation layer rather than to the projection kernel. The memory scale τ was set by a pre-registered sweep over {7, 14, 30 days} rather than a single a priori choice; the enrichment increases monotonically with τ in both records while the flagged-time fraction stays near 4%, which argues against alert-budget dilution as the explanation. We report the full sweep, state explicitly what the method does not claim, and note that a confirmatory study with τ fixed before data exposure on an independent record is the appropriate next step.

Keywords: cascading failures; transmission outages; conditional severity; point processes; non-Markovian memory; early warning; BPA outage data.

1. Introduction

Cascading outages are a principal mechanism of large blackouts, and their statistics — heavy-tailed size distributions, criticality, and propagation — have been characterized extensively from both models and utility data [1–4]. A substantial part of that literature addresses propagation (how outages beget outages once a cascade is underway) and occurrence (when cascades happen). Occurrence, in particular, is strongly driven by exogenous forcing: for records such as the BPA public outage data, weather may directly cause a large share of outages, and the proportion of weather-driven versus interaction-driven events is not known [5]. This context, emphasized by the curator of the dataset, cautions against interpreting occurrence predictability as evidence about system state.

This paper deliberately targets a different quantity: conditional severity. The question is not will a cascade occur in the next hours? but if a cascade occurs now, is the system in a state in which it is more likely to become large? The distinction matters operationally and scientifically. Operationally, severity-conditioned risk is what escalation procedures, reserve posture, and maintenance deferral decisions respond to. Scientifically, it separates two sources of predictability that occurrence metrics conflate: the exogenous arrival of disturbances (plausibly weather-dominated) and the internal disposition of the system to amplify a disturbance once it arrives.

Our hypothesis is that this disposition is a property of the system's recent-to-medium history, not only of its instantaneous or recent activity. Concretely, we test whether a strictly causal projection of the outage stream — one that (i) measures the decoupling between observed activity and the system's own causally expected activity, (ii) accumulates that decoupling with genuine memory over multiple weeks, and (iii) compares the accumulation against a threshold that itself contracts as excess accumulates — identifies periods of elevated conditional severity that a memoryless (Markovian) intensity baseline does not.

The projection used here has a general, domain-independent formulation; this paper, however, is self-contained: every quantity used in the analysis is defined below from first principles as a causal, memory-bearing state estimator for event streams, and no reported result depends on accepting any broader theoretical claim. Readers interested in the general formulation may consult [6, 7]; nothing in that material is required to evaluate what follows.

Three design commitments shape everything that follows. Strict causality: every quantity at time t is a function of the stream before t ; truncating the record must leave all earlier values unchanged, and this holds for the baselines as well (a future-leaking baseline discovered during development was removed before any comparative evaluation). Transparent parameter selection: the memory scale τ was chosen by an explicit, reported sweep over a small pre-specified grid rather than a single value fixed by assumption; the sweep and its outcome are reported in full in §4.1, not folded silently into a single number. Negative results retained: results where the baseline wins (occurrence forecasting, §4.1) and a measurement failure discovered during the NYISO analysis, together with the ablation that localizes it, are reported as found.

2. Data

2.1 BPA transmission outage record

We use the cleaned version of the Bonneville Power Administration public transmission outage record, 1999–2017, prepared and documented by I. Dobson [5] (CFAREADBPA-10 cleaning and documentation). Following standard practice for cascading analysis of this record, only automatic outages are used, yielding 14,258 events. Outages are grouped into cascades using 1-hour gap grouping, matching the

repository implementation (aptadynamic_eg.omega.cascades, GAP_S = 3600 s); cascade size is the number of automatic outages in a group. Large cascades are defined as size ≥ 4 .

Two properties of this record must frame all interpretation. First, as noted by its curator, the record includes — and may be dominated by — outages caused directly by weather; the proportion of cascading versus weather-driven outages is not known [5]. We therefore do not interpret any result below as evidence about cascading mechanisms; we interpret them as statements about the statistical structure of the outage stream. Second, the record is a stream of events with timestamps and (anonymized) equipment identifiers; no system topology, loading, or meteorological covariates are used anywhere in the analysis.

2.2 NYISO record

As a second, independent record we use processed NYISO forced-outage data (2008–2021, 9,600 events) from a published web source, shared in processed form by I. Dobson, who notes it is of lower quality than the BPA record. Event density is substantially lower and the processing history differs; §4.2 discusses the consequences, including a duration field in the raw source explicitly flagged as unreliable during ingestion.

2.3 Sensitivity and attribution

All bus and equipment identifiers are anonymized in every output generated by the analysis pipeline, and no complete or potentially sensitive details of particular cascading sequences are reported here or in the repository. Per BPA's request, the attribution statement in the Disclaimer applies to this work in its entirety.

3. Method

3.1 Observable stream

Time is binned hourly. The primary observable $\omega(t)$ is the raw outage count per bin (no pre-normalization is applied at this stage). Everything from §3.2 onward, including the normalization implicit in Eq. 1, is identical in form across datasets; only the input stream (BPA vs. NYISO) differs between the two applications.

3.2 Causal expectation

The system's own expected activity $\hat{\omega}(t)$ is the strictly causal conditional mean of ω given a seasonal context (calendar month $\times$ hour of day): at each bin, $\hat{\omega}$ is the running mean of all past bins sharing that context, once the context has been observed at least $n_{\min} = 10$ times; before that, the running global mean is used once at least 720 past bins exist; earlier bins are treated as warm-up and excluded from evaluation. By construction, truncating the stream at t leaves $\hat{\omega}(s)$ unchanged for all $s \leq t$.

3.3 Decoupling

The instantaneous decoupling between observed and expected activity is

$$\Delta(t) = |\omega(t) - \hat{\omega}(t)| / (\hat{\omega}(t) + 1). \quad (1)$$

Δ is deliberately not a normalized activity measure: it is small when the system behaves as its own history predicts (even during seasonally busy periods) and large when behavior departs from that

prediction. §4.2 documents, as a controlled ablation, what happens when this distinction is inadvertently lost.

3.4 Memory accumulator

Decoupling is accumulated with an exponential causal kernel,

$$\Xi(t) = \sum_{s < t} K(t-s) \Delta(s), \quad K(u) = (1-a) a^u, \quad a = e^{-1/\tau}, \quad (2)$$

with memory scale τ selected by the sweep reported in §4.1 (final value: $\tau = 720$ h, 30 days). Ξ is the non-Markovian component of the method: two intervals with identical present activity but different multi-week histories carry different Ξ .

3.5 Permissivity and endogenous threshold

A permissivity state $\lambda(t) \in [\lambda_{\min}, \lambda_{\text{eq}}]$ tracks the system's remaining absorption capacity. It is eroded by excess accumulation over the current threshold and recovers toward equilibrium at a bounded rate:

$$\lambda(t+1) = \text{clip}(\lambda(t) - \kappa(\Xi(t) - \Theta(t))^+ + r(\lambda_{\text{eq}} - \lambda(t)), \lambda_{\min}, \lambda_{\text{eq}}), \quad (3)$$

$$\Theta(t) = \theta_s \cdot \lambda(t). \quad (4)$$

Two structural asymmetries are enforced. The threshold Θ is endogenous — it contracts as excess accumulates, rather than being a fixed cutoff on any score. And recovery is one-way: quiet periods restore λ but never reduce Ξ , which decays only through its own kernel (2).

The derived quantities are the margin $M(t) = \Theta(t) - \Xi(t)$ and the smoothed margin trend $G(t) = D^+ \langle M \rangle_w$ (one-sided derivative of a $w = 24$ h rolling mean).

3.6 Flagged periods

A bin is flagged when the system is operational, the margin is still nonnegative, and the margin is being consumed:

$$\text{flag}(t) = 1 \iff \omega(t) > 0 \wedge M(t) \geq 0 \wedge G(t) < 0. \quad (5)$$

The flag is thus explicitly not an exceedance alarm: it fires while the state still looks acceptable by level, on the basis of its direction. This is the quantity whose severity-conditioning power the study evaluates.

3.7 Parameters

Structural parameters not subject to the sweep were fixed prior to, and are identical across, both datasets: $\lambda_{\text{eq}} = 1$; $r = 0.005$ per bin; $\lambda_{\min} = 0.1$; $\theta_s = 2.0$; $w = 24$ h; $\kappa = 0.05$. The observation layer (§3.1–3.2) has two integer warm-up constants (10, 720) fixed a priori. The memory scale τ is the single parameter selected by the reported sweep described in §4.1; it is identical across BPA and NYISO once selected ($\tau = 720$ h for all results reported outside §4.1's sweep table).

3.8 Baselines

The primary comparator is a causal trailing-intensity baseline: the outage intensity over a trailing 12-hour window, shifted by one bin to preserve strict causality, thresholded to produce the same alert budget (fraction of flagged bins) as the projection, so that comparisons are at matched operating points. This baseline is the natural memoryless (Markovian) competitor: it answers "has the system been active lately?" A future-leaking variant (using centered or forward-looking statistics) was implemented by

mistake early in development, detected by a truncation-invariance test, and removed before any comparative evaluation; only strictly causal baselines are reported.

3.9 Evaluation protocol

Conditional severity. For each signal (projection flag; budget-matched baseline flag), we estimate $P(\text{size} \geq 4 \mid \text{cascade occurs, flag state})$ and report the enrichment ratio between flagged and unflagged states. Significance is assessed with a label-shuffle permutation test (1,000 permutations of cascade severity labels across cascade occurrences), reporting p-values and the null distribution's 95th percentile.

Overlap partition. To test whether the projection merely proxies recent activity, cascades are partitioned into four classes by the flag state of the two signals at cascade onset (both / projection-only / baseline-only / neither), and conditional severity is estimated per class.

Occurrence forecasting (control question). For horizons of 6–48 h we evaluate whether each signal predicts cascade occurrence, to establish which questions the projection does and does not help with.

External consistency. As a validation check — not a calibration target — the pipeline estimates the Zipf exponent of the cascade-size distribution for comparison with published values for this record.

4. Results

4.1 Memory-scale sweep and conditional severity (BPA, NYISO)

The memory scale τ was selected by an explicit, pre-specified sweep over $\tau \in \{168, 336, 720\}$ h (1, 2, and 4 weeks), evaluated on both records under the identical protocol of §3.9. We report the full sweep rather than a single chosen value, because the choice of τ materially affects the headline enrichment ratio and we want that dependence visible rather than folded into one number.

τ (memory scale)	BPA enrichment	NYISO enrichment	Flagged-time fraction (both records)
168 h (7 days)	16.0	1.90	$\approx 4.2\text{--}4.6\%$
336 h (14 days)	23.4	2.12	$\approx 4.1\text{--}4.5\%$
720 h (30 days)	28.8	2.44	$\approx 4.1\text{--}4.6\%$

Table 1. Conditional-severity enrichment ratio, $P(\text{size} \geq 4 \mid \text{flagged}) / P(\text{size} \geq 4 \mid \text{unflagged})$, as a function of memory scale τ , on both records. The flagged-time fraction (alert budget) stays within a narrow band across the full range of τ , which argues against alert-budget dilution or inflation as the explanation for the increase in enrichment with τ .

The enrichment increases monotonically with τ in both records over this range, and we adopt $\tau = 720$ h (30 days) as the operating value for all results reported below and in §4.2–4.4, since it is both the largest value tested and the one with the clearest separation from the causal Markovian baseline. We did not test values beyond 720 h; whether the enrichment continues to increase, plateaus, or eventually degrades at longer memory scales is accordingly not established by this study, and we list it as an open question in §5.

At $\tau = 720$ h, conditioning on cascade occurrence, the probability of a large cascade is $P(\text{size} \geq 4 \mid \text{cascade, flagged}) = 0.096$ versus $P(\text{size} \geq 4 \mid \text{cascade, unflagged}) = 0.003$, an enrichment ratio of ≈ 28.8 (permutation $p < 0.001$; null 95th percentile ≈ 1.17). At the same alert budget, the trailing-intensity baseline achieves an enrichment of ≈ 3.2 — a figure that, unlike the projection's, does not depend on τ . The gap between 28.8 and 3.2 is the paper's central quantitative claim: conditional severity carries

structure that recent intensity alone does not capture, and that structure is accessible to a strictly causal, memory-bearing estimator.

Occurrence forecasting remains Markovian at every memory scale tested. Across 6–48 h horizons, the trailing-intensity baseline consistently outperforms the projection for forecasting whether cascades occur:

Horizon	Projection enrichment (occurrence)	Trailing-intensity enrichment (occurrence)
6 h	2.17	2.67
12 h	2.14	2.56
24 h	1.73	2.45
48 h	1.70	1.91

Table 2. Occurrence-forecasting enrichment (predicting whether a cascade happens, not its size) by horizon, BPA. The baseline outperforms the projection at every horizon tested.

This is the expected outcome if occurrence is dominated by exogenous forcing (e.g., weather) rather than by system state, and is consistent with the curator's caution about this record [5]. We report it as a boundary of the method's applicability, not as an anomaly.

4.2 Ablation: decoupling versus raw intensity

The distinction between Δ as structural decoupling (Eq. 1) and Δ as a proxy for raw normalized intensity is not merely definitional: replacing one with the other changes the sign of the NYISO result. We report this as a controlled ablation rather than an incidental bug fix, because it bears directly on what kind of information the method extracts.

Record	$\Delta \approx$ raw intensity (degenerate)	$\Delta =$ decoupling, Eq. 1 (reported)
BPA	4.95	28.8
NYISO	0.55 (inverted)	2.44

Table 3. Conditional-severity enrichment ratio by realization of Δ . The degenerate realization collapses the distinction between Δ and $\hat{\omega}$ (i.e., $\hat{O}$ held constant), so Ξ accumulates raw activity rather than structural deviation.

With the degenerate Δ , BPA still yields a positive enrichment (4.95) because its higher event density partially masks the degeneration, but NYISO inverts below chance (0.55): in a sparser record, accumulating raw intensity rather than deviation from expectation actively misleads the conditioning. Restoring Eq. 1's intended semantics — with the projection kernel and its parameters untouched — raises BPA to 28.75 and turns NYISO positive (2.44). The failure therefore localized to the observation layer, not to the projection kernel; because the two are architecturally separated in the pipeline, the fix did not require touching the accumulator, permissivity, or threshold logic at all.

Under a strict pre-registration standard, this implies the NYISO analysis should be read as exploratory rather than confirmatory for this reason as well as for the τ sweep of §4.1: both the observation layer and the memory scale were adjusted after some outcome exposure during development. A confirmatory replication on an independent record, with both choices fixed in advance, is future work, and we flag this explicitly rather than let it pass silently.

4.3 Disentangling memory from recent activity

The overlap partition addresses the natural objection that flagged periods are simply busy periods. At $\tau = 168$ h, 336 h, 720 h, partitioning BPA cascades by the intersection of the projection flag and the budget-matched Markovian alert at cascade onset gives:

Partition (BPA, $\tau = 168$ h)	n cascades	P(size ≥ 4)
Both flagged	627	0.175
Projection only	5,295	0.081
Baseline only	3	0.000
Neither flagged	2803	0.006

Partition (BPA, $\tau = 336$ h)	n cascades	P(size ≥ 4)
Both flagged	627	0.175
Projection only	5,134	0.085
Baseline only	0	0.000
Neither flagged	3,007	0.004

Partition (BPA, $\tau = 720$ h)	n cascades	P(size ≥ 4)
Both flagged	630	0.175
Projection only	5,091	0.086
Baseline only	0	— (empty)
Neither flagged	3,007	0.003

Table 4. Independence partition, BPA, $\tau = 720$ h. The baseline-only cell is empty at this memory scale: every budget-matched Markovian alert falls inside a projection-flagged period, while 89% of projection-flagged cascades (5,091 of 5,721) carry no baseline alert.

For cascades occurring in bins flagged only by the projection, conditional severity remains 0.086, versus 0.003 where neither signal is flagged — a ratio of ≈ 29 with the trailing-intensity signal entirely absent. We note the empty baseline-only cell explicitly rather than omit it: at this memory scale, the Markovian baseline no longer identifies any elevated-risk period that the projection misses, which is a stronger, but also a more scale-dependent, version of the independence claim than at shorter τ . Tables 4a–4b report the same partition at $\tau = 168$ h and 336 h, where the baseline-only cell is non-empty but small ($n = 3$ in both cases): Table 4a. Independence partition, BPA, $\tau = 168$ h.

4.4 External consistency

The pipeline's estimate of the Zipf exponent for BPA cascade sizes is ≈ 2.99 . Published estimates for related quantities on this record cluster in the range of roughly 2.9–3.0 depending on the exact quantity fit (cascade size versus number of generations per cascade) and the fitting method (log-log rank regression versus maximum-likelihood with KS-optimized cutoff). We emphasize the direction of inference: the exponent was computed as a downstream check that the grouping and filtering reproduce statistics of the expected shape for this kind of record; no component of the projection was calibrated to it. We also note the curator's assessment that a Zipf description of this record is simpler and preferable to increasing propagation in a branching process [5]; nothing in our method presumes either description.

4.5 NYISO: a measurement failure and a weaker positive result

As detailed in the ablation (§4.2), applying the projection to the NYISO record with a degenerate observation layer initially produced an enrichment of 0.55 — worse than chance. After correcting the observation layer to restore Eq. 1's intended semantics, and after the τ sweep of §4.1, the enrichment at $\tau = 720$ h is 2.44, exceeding the permutation null (95th percentile ≈ 1.26). The NYISO source data also carries a duration field documented during ingestion as unreliable; it is not used in the projection (which depends only on event counts per bin), but it further supports treating this record as lower-quality than BPA, consistent with its curator's characterization.

Our working interpretation of the weaker NYISO enrichment relative to BPA (2.44 vs 28.8) is informational: with roughly two-thirds fewer events over a longer span, and a processing history that differs from the BPA record, the stream simply carries less usable history per unit time for a memory-bearing estimator. We consider this interpretation plausible but not established.

5. Discussion

What the results say. On the best available public cascading record, periods identified by a strictly causal projection — one that accumulates the system's departure from its own expected behavior over a multi-week memory and compares it against a history-contracted threshold — carry, at the memory scale selected in §4.1, a ≈ 29 -fold elevation of conditional cascade severity, of which only a ≈ 3 -fold elevation is attributable to recent activity. Severity disposition, unlike occurrence, appears to be a property of history, though the exact magnitude of that elevation depends on the memory scale chosen, as Table 1 makes explicit.

What the weather caveat does to interpretation. Because the BPA record may be dominated by weather-caused outages in unknown proportion [5], we do not claim that flagged periods correspond to any specific physical mechanism (protection-system stress, loading, maintenance state, or weather clustering itself). The honest reading is narrower and still useful: whatever mixture of mechanisms generates this stream, its severity-relevant state is partially encoded in the stream's own history, in a form that a memoryless intensity statistic does not extract. Notably, this reading is compatible with weather dominance: severe-weather epochs themselves have temporal structure that a memory-bearing estimator can exploit, and disentangling "system disposition" from "environmental disposition" would require covariates that this analysis deliberately does not use.

Relation to propagation and Zipf descriptions. The method is agnostic between branching-process and Zipf descriptions of cascade sizes [5, 8]: it conditions on occurrence and asks about size, whatever generates the size distribution. Its contribution is orthogonal — a when (in state, not in time) to the literature's how large.

Limitations. (i) Single fully validated record; the NYISO replication is positive but exploratory, both because its observation layer was revised after outcome exposure (§4.2) and because the operating τ was selected by a sweep evaluated on both records (§4.1). (ii) No covariates: the design isolates stream-intrinsic information at the cost of mechanistic attribution. (iii) The 1-hour grouping and size-4 severity cutoff follow prior practice on this record but remain conventions. (iv) The memory scale τ was selected from a small grid {168, 336, 720} h by observing enrichment on the same two records subsequently used to report results; this is disclosed in full in Table 1 rather than presented as a value fixed in advance, and it means the headline figures in the abstract and §4.1 should be read as the outcome of that selection, not as validation of a pre-registered choice. A study on a fresh record with τ and all other choices pre-registered before data exposure is the appropriate confirmatory standard, and is planned.

What this study does not claim. The method does not forecast cascade occurrence (the memoryless baseline is better at that, §4.1); does not predict the timing of large cascades; does not model grid physics, topology, or weather; and does not, on this evidence, establish mechanism. Claims beyond conditional-severity discrimination on these records, at the memory scale reported, should not be attributed to it.

6. Conclusion

Conditioning on occurrence separates what is predictable about cascading outages into two different questions with two different answers. Whether cascades occur looks Markovian and plausibly exogenous, at every memory scale tested. How severe they become, given that they occur, is substantially non-Markovian: on the BPA record, a strictly causal projection with a multi-week memory (selected by the sweep in Table 1) and an endogenous threshold identifies periods with ≈ 29 -fold elevated conditional severity, roughly nine-fold beyond a matched-budget recent-intensity baseline, with the excess persisting in periods the baseline does not flag at all at this memory scale.

If replicated under pre-registered conditions on independent records — with the memory scale and observation-layer definition both fixed before data exposure — severity-conditioned, memory-bearing state estimation would offer transmission operators a complementary quantity to occurrence risk: not when the next disturbance arrives, but whether the system is currently disposed to amplify it.

Data and code availability

The full analysis pipeline, configuration, and generated (anonymized) outputs are public at <https://github.com/gacj0901/Aptadynamic-Electrical-Grid>. The cleaned BPA record and its cleaning documentation (CFAREADBPA-10) are available from I. Dobson's public repository [5].

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Disclaimer

The analysis and any conclusions are strictly those of the authors and not of Bonneville Power Administration.

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Manuscript prepared for assessment by domain experts prior to preprint submission. Comments on the conditional-severity framing, the causal baseline construction, the memory-scale sweep, and the interpretation of the BPA–NYISO difference are particularly welcome.